

Artificial Intelligence and Data Science

CSA1008 – Software Engineering

Assignment Title: CO5-AT1-Research Paper/Literature Review Survey

Case Study Title: Smart Waste Collection System

(SDG 11: Sustainable Cities and Communities)

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1. Introduction

Municipal waste collection is commonly carried out according to fixed schedules. However, the amount of waste generated in different locations is not always the same. Some bins may become full before the scheduled collection, while collection vehicles may visit bins that are still mostly empty.

This creates problems such as overflowing waste, poor hygiene, unnecessary vehicle trips, increased fuel consumption and higher operational costs.

The proposed Smart Waste Collection System solves this problem by using IoT sensors, cloud computing, GPS and route optimization. Smart bins monitor their waste levels and send the information to a central system. The system identifies bins requiring collection and generates an optimized route for collection vehicles.

2. Problem Statement

The existing fixed-schedule waste collection system does not consider the actual fill level of waste bins.

Major problems

- Overflowing bins may not be detected immediately.
- Empty or partially filled bins may be collected unnecessarily.
- Collection vehicles travel inefficient routes.
- Fuel and maintenance costs increase.
- Municipal authorities have limited real-time information.
- Manual monitoring can result in errors.
- It is difficult to track completed and missed collections.

Therefore, a software system is required to monitor bins, optimize collection routes and provide real-time operational information.

3. Objectives

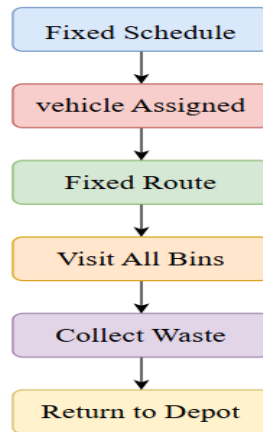
The proposed system aims to:

1. Monitor waste-bin levels in real time.
2. Detect bins approaching their capacity.
3. Generate alerts for critical bins.
4. Optimize waste collection routes.
5. Reduce unnecessary vehicle travel.
6. Reduce fuel and operational costs.
7. Track collection vehicles using GPS.
8. Provide a centralized municipal dashboard.

9. Maintain historical collection data.
10. Improve the efficiency and sustainability of waste management.

4. Existing System and Limitations

Existing System



Limitations

Existing Approach	Problem
Fixed collection schedule	Does not consider actual waste level
Manual monitoring	Time-consuming and error-prone
Fixed routes	May result in unnecessary travel
No real-time monitoring	Overflowing bins may remain unnoticed
Limited vehicle tracking	Difficult to monitor operations
Manual records	Difficult to analyse historical data

5. Proposed Smart Waste Collection System

The proposed system changes waste collection from a fixed-schedule approach to a demand-based approach.

Each smart bin contains a sensor that measures its approximate fill level. The information is transmitted to the central server. When a bin reaches a predefined threshold, the system generates an alert and considers the bin for collection.

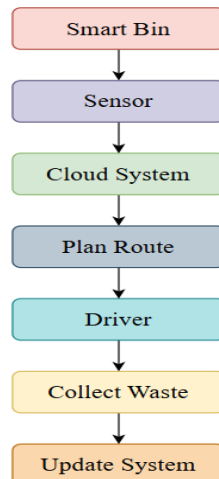
The route optimization module then considers:

- Fill level
- Bin priority
- Vehicle capacity
- Vehicle location

- Distance
- Traffic conditions
- Collection time

The optimized route is sent to the driver's mobile application.

Proposed Process



6. System Requirements

6.1 Functional Requirements

ID	Requirement
FR1	Register and manage waste bins
FR2	Receive sensor readings
FR3	Display bin fill levels
FR4	Generate overflow alerts
FR5	Register vehicles and drivers
FR6	Track collection vehicles
FR7	Generate optimized routes
FR8	Assign routes to drivers
FR9	Record completed collections
FR10	Generate reports and analytics

6.2 Non-Functional Requirements

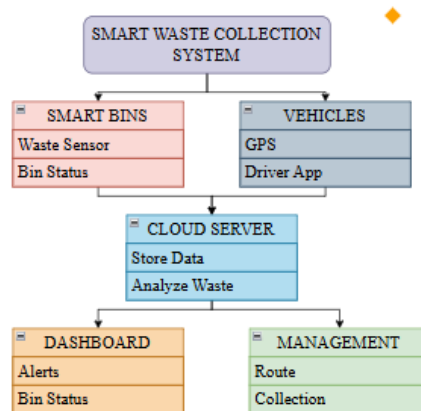
Requirement	Description
Performance	System should process sensor data quickly

Reliability	Data should not be lost during temporary failures
Security	User and system data must be protected
Scalability	System should support more bins and vehicles
Usability	Dashboard should be easy to understand
Availability	System should remain accessible during collection operations
Maintainability	System should be modular and easy to update

7. System Design

7.1 System Architecture

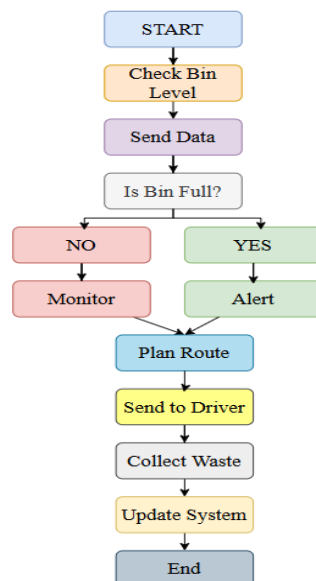
For your report, create this properly in **draw.io**.



Why this architecture?

It separates the system into independent layers, making it easier to develop, test, maintain and upgrade.

7.2 System Flowchart



8. Main Modules

1. User Management

Manages:

- Administrator accounts
- Supervisor accounts
- Driver accounts
- Login and authentication

2. Smart Bin Management

Manages:

- Bin registration
- Bin location
- Sensor information
- Fill level
- Bin status

3. Route Optimization

Selects bins requiring collection and generates an efficient route based on distance, priority and vehicle capacity.

4. Vehicle Management

Maintains:

- Vehicle details
- Driver assignment
- Vehicle capacity
- GPS location
- Vehicle availability

5. Alert Management

Generates alerts for:

- High fill levels
- Overflow risk
- Sensor failure
- Missed collection

6. Dashboard and Analytics

Displays:

- Total bins
- Critical bins
- Active vehicles
- Completed collections
- Missed collections
- Historical statistics

9. Use Case Diagram

Actors

- Administrator
- Supervisor
- Driver
- Citizen
- Smart Bin

10. Database Design

The system can use **MySQL or PostgreSQL**.

Main Tables

Table	Important Fields
Users	user_id, name, email, role
Bins	bin_id, location, capacity, sensor_id, status
Sensor_Readings	reading_id, bin_id, fill_level, timestamp
Vehicles	vehicle_id, registration_no, capacity, status
Drivers	driver_id, user_id, vehicle_id
Routes	route_id, vehicle_id, date, status
Route_Stops	route_id, bin_id, sequence
Collections	collection_id, bin_id, driver_id, timestamp
Alerts	alert_id, bin_id, alert_type, priority

11. Technology Stack

Component	Technology
Frontend	React.js

Mobile App	React Native
Backend	Python Fast API
Database	MySQL / PostgreSQL
IoT Controller	ESP32
Sensor	Ultrasonic Sensor
Communication	MQTT / HTTP
Cloud	Firebase / AWS
Maps	OpenStreetMap / Maps API
Route Optimization	Google OR-Tools
Development	VS Code
Version Control	GitHub
API Testing	Postman
Diagrams	draw.io
UI Prototype	Figma

You do not need to actually implement every technology. This table represents the proposed technology architecture.

12. Agile Development and Testing

The system can be developed using Agile Scrum.

Sprint	Work
Sprint 1	Requirements and system design
Sprint 2	User and bin management
Sprint 3	IoT monitoring
Sprint 4	Route optimization
Sprint 5	Driver application
Sprint 6	Dashboard and analytics
Sprint 7	Integration and testing
Sprint 8	Deployment and maintenance

Testing

Test Case	Expected Result
Valid login	User successfully logs in

Invalid login	Error displayed
Bin reaches 90%	Critical alert generated
Bin below threshold	No urgent collection
Route generated	Optimized route displayed
Collection completed	Status updated
Sensor failure	Failure alert generated
Unauthorized access	Access denied

13. Security and Responsible Software Engineering

Security should be included from the beginning of development.

Security

- Password hashing.
- Role-based access control.
- HTTPS communication.
- API authentication.
- Input validation.
- Database backups.
- Secure IoT credentials.
- Audit logs.

Responsible Software Engineering

The system should also consider its social impact.

- Collect only necessary user information.
- Protect driver and administrator data.
- Provide accessible interfaces.
- Maintain human supervision over automated decisions.
- Allow supervisors to override routes during emergencies.
- Provide reliable fallback procedures when sensors fail.
- Monitor the environmental impact of the system.

14. Benefits, Challenges and Future Enhancements

Benefits

Area	Benefit
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Operational	Efficient routes and better vehicle utilization
Economic	Lower fuel and operating costs
Environmental	Reduced unnecessary vehicle travel
Social	Cleaner and healthier surroundings
Management	Real-time monitoring and analytics

Challenges

- Initial sensor installation cost.
- Sensor maintenance.
- Internet connectivity problems.
- Incorrect sensor readings.
- Cybersecurity risks.
- Training requirements.
- Dependence on accurate location and traffic data.

Research Gap

Current smart waste systems demonstrate the usefulness of IoT monitoring and route optimization, but a major challenge is achieving consistent performance in real-world environments. Some approaches focus mainly on sensor monitoring, while others focus mainly on route optimization. A stronger solution should combine:

Real-Time Monitoring + Route Optimization + Vehicle Capacity + GPS + Predictive Analysis + Human Decision-Making

Therefore, the proposed system addresses the gap by treating waste collection as a continuous and adaptive software process, rather than simply monitoring bin levels.

Future Enhancements

1. AI-based prediction of future bin fill levels.
2. Traffic-aware dynamic route generation.
3. Predictive maintenance of sensors and vehicles.
4. Carbon-emission monitoring.
5. Citizen mobile application.
6. AI-based waste classification.
7. Multi-objective optimization considering distance, time, fuel and emissions.

15. Conclusion

The proposed Smart Waste Collection System provides an efficient software engineering solution to the limitations of fixed-schedule garbage collection. By combining IoT sensors, cloud computing, GPS, route optimization and data analytics, the system can monitor waste levels and dynamically determine which bins require collection.

Instead of sending vehicles to every bin according to a fixed schedule, the proposed system prioritizes bins based on their actual condition and generates suitable collection routes. This can help reduce unnecessary travel, fuel consumption, operational costs and overflowing waste.

The solution also demonstrates responsible software engineering through security, privacy, human oversight, scalability, maintainability and environmental considerations. The system can initially be implemented as a pilot project in selected areas and gradually expanded across the city. Future integration of AI-based prediction and dynamic traffic-aware routing can make the platform more intelligent and adaptive.

Thus, the proposed system transforms traditional waste collection into a data-driven, responsive and sustainable urban service.